What Shapes Political Discourse? Party Affiliation vs. Candidate Profession in French Electoral Manifestos*

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Abstract

This study examines whether political discourse in French legislative campaign materials is shaped more by party affiliation or by candidate socio-professional background. We analyze a cleaned corpus of 3,596 campaign documents from the 1981, 1988, and 1993 legislative elections. After preprocessing the texts and harmonizing profession and party labels, we use Latent Dirichlet Allocation (LDA) to extract thematic structure and BERTopic as a complementary robustness check.

The topics capture major dimensions of campaign discourse, including left–right conflict, social and economic policy, environmental issues, and national political rhetoric. We then regress document-level topic proportions on party, profession, and election year. The results show that party affiliation is by far the strongest correlate of topic usage: in the baseline specification, party explains about 0.407 of average partial R^2, compared with only 0.005 for profession. This result remains stable across alternative specifications and robustness checks.

Overall, the findings suggest that campaign discourse in this corpus is structured primarily by party-level dynamics rather than individual candidate backgrounds.

1 Introduction

Understanding what shapes political discourse during electoral campaigns is a central question in political science and computational social science. Campaign manifestos and candidate leaflets provide direct evidence on how political actors frame issues, communicate priorities, and appeal to voters. Yet an important empirical question remains open: are these messages driven primarily by collective party structures, or do they also reflect individual candidate characteristics such as socio-professional background? [2, 3]

This project studies that question using a cleaned corpus of 3,596 French legislative campaign documents from the 1981, 1988, and 1993 elections, drawn from the CEVIPOF political archives [7]. The final analytical sample is obtained after substantial cleaning of archival texts, including the removal of very short documents, exclusion of observations with missing party or profession labels, and harmonization of highly granular metadata. In particular, the profession field is aggregated into broad categories to make comparisons tractable.

To measure campaign discourse, we use Latent Dirichlet Allocation (LDA) [1] to estimate document-topic mixtures that capture the main thematic dimensions of the corpus. We then use these topic proportions as outcome variables in regressions on party affiliation, profession, and election year. BERTopic [4] is used as a complementary robustness check for topic interpretation.

Our results show that party affiliation is by far the strongest correlate of topic usage. In the baseline specification, the average partial R^2 attributable to party is 0.407, compared with only 0.005 for profession. This pattern remains stable across alternative topic numbers, preprocessing choices,

*Code is available at https://github.com/sarakpyny/french-political-text-analysis.

profession regroupings, and sample restrictions. Additional party-level analysis also shows that several parties display strong internal cohesion in topic space.

This study contributes to the literature in two ways. It shows that French legislative campaign discourse in this period is structured much more by party-level organization than by candidate socio-professional background. On the other hand, it illustrates how topic modeling, variance decomposition, and robustness analysis can be combined to compare collective and individual determinants of political messaging.

2 Related Work

The use of textual data to study political behavior has expanded rapidly with the development of computational methods in political science and economics. Foundational work has emphasized both the promise of automated text analysis for extracting structure from large corpora and the methodological challenges of interpretation, measurement, and validation [2, 3]. These approaches are now widely used to study speeches, manifestos, parliamentary debate, and campaign materials.

Topic modeling has become a standard tool for uncovering latent thematic structure in political texts. In particular, Latent Dirichlet Allocation (LDA) provides a probabilistic framework in which documents are represented as mixtures of topics and topics as distributions over words [1]. LDA has been widely used to study ideology, policy agendas, and issue competition. Related work has also extended topic models to incorporate document-level metadata, for example through Structural Topic Models [6]. More recently, embedding-based approaches such as BERTopic have offered complementary ways to capture semantic similarity in noisy or heterogeneous corpora [4].

Beyond topic discovery itself, an important strand of research examines the political determinants of discourse. Existing work shows that party strategies, issue competition, and responsiveness to public opinion shape the salience and positioning of political messages [5]. At the same time, broader debates on political personalization suggest that individual candidate characteristics may also matter for communication. This raises an important empirical question: when campaign texts vary across candidates, how much of that variation reflects party-level organization, and how much reflects individual background?

This study contributes to that literature by combining topic modeling with regression-based variance decomposition to compare the relative importance of party affiliation and candidate profession in structuring campaign discourse. Rather than using topic models only for descriptive exploration, we use document-topic proportions as outcomes in a systematic comparison of collective and individual determinants of political messaging. In doing so, this study speaks both to the text-as-data literature and to broader debates on partisan coordination and candidate-level personalization in electoral communication.

3 Data

3.1 Corpus Description

Our analysis is based on a corpus French legislative campaign documents covering the 1981, 1988, and 1993 elections. These documents consist of candidate leaflets and campaign materials collected from the CEVIPOF political archives at Sciences Po [7]. They provide rich textual evidence on how candidates present political positions, policy priorities, and ideological messages to voters.

Each document is matched with candidate-level metadata, including party affiliation, profession, gender, age, and constituency. Because the raw metadata are highly granular, both party and profession labels require harmonization before analysis. In particular, the profession field contains 2,030 distinct raw labels, which makes direct analysis too sparse. We therefore group professions into 10 broad categories such as, *Education, Law, Health, Agriculture, Business, Technical, Worker, Media, Political, Other*, using a rule-based harmonization procedure.

Party labels are also highly fragmented. After excluding observations with missing party labels, we retain party labels represented by more than 100 documents and collapse the remaining smaller groups into an "Other" category. One spelling inconsistency ("Liste entente populaire et nationale"

vs. "Liste d'entente populaire et nationale") is then corrected, yielding 12 party categories in the final analytical sample.

3.2 Text Preprocessing

The archival documents contain substantial noise, including formatting artifacts, OCR errors, and repeated institutional headers. To improve text quality, we apply several preprocessing steps.

First, we normalize the text by lowercasing and removing punctuation, numbers, and non-alphabetic tokens. We then lemmatize the corpus to reduce words to their base forms. Standard French stopwords are removed, together with corpus-specific stopwords designed to filter out archival markers, election-formatting terms, and recurrent OCR artifacts (e.g., *CEVIPOF*, *circonscription*, and residual German OCR words). We use this preprocessing to reduce noise, standardize the text, and preserve meaningful political content for analysis.

To construct the document-term matrix used for topic modeling, we apply a `CountVectorizer` with a minimum document frequency of 20, a maximum document frequency of 95%, and a restriction to alphabetic tokens of at least three characters. These thresholds reduce sparsity and residual noise by excluding very rare terms, OCR artifacts, and short tokens, while also removing overly common words with limited topical discrimination. We use raw term counts rather than TF-IDF because LDA is a probabilistic generative model defined over observed word counts, making count-based features more appropriate for estimation and interpretation.

We also apply sample restrictions before modeling. Starting from the OCR-linked legislative documents for 1981, 1988, and 1993, we retain only texts longer than 100 words and exclude observations with missing profession ("non mentionné") or missing party labels. These restrictions improve comparability across candidates and reduce noise from documents with limited textual content. The resulting cleaned dataset contains 3,596 documents.

3.3 Descriptive Statistics

The final dataset contains 3,596 documents with substantial variation in both party affiliation and socio-professional background. The largest party groups are "Other" (864 documents), "Liste d'entente populaire et nationale" (450), and "Entente des écologistes" (421). Profession categories are also heterogeneous, with "Other" (1,454), "Education" (617), and "Health" (303) being the most common groups.

Documents are relatively long campaign texts, with an average length of about 637 words and a median of 555 words. Frequent terms in the cleaned corpus include words such as *national, politique, france, social, and emploi*, pointing to recurring concerns around national politics, social issues, and employment.

These descriptive patterns indicate that the corpus contains meaningful political variation while remaining sufficiently structured for topic modeling and regression analysis.

4 Methodology

Our empirical strategy combines unsupervised topic modeling with regression analysis to study the determinants of political discourse. We first extract latent topics from campaign documents, and then analyze how topic usage varies with candidate characteristics. This design allows us to compare whether campaign discourse is structured more strongly by collective party affiliation or by individual socio-professional background.

4.1 Topic Modeling

To uncover the thematic structure of the corpus, we apply Latent Dirichlet Allocation (LDA) [1]. LDA is a generative probabilistic model in which each document is represented as a mixture of topics, and each topic is a distribution over words. Formally, for each document d, the model estimates a topic distribution θ_d, where each element gives the proportion of a given topic in the document.

We construct a document-term matrix using a bag-of-words representation based on `CountVectorizer` and a filtered vocabulary. We use count-based features because standard LDA is defined over observed word counts, making raw counts more appropriate for estimation and interpretation than TF-IDF weighted inputs. The LDA model is estimated using the batch learning procedure implemented in `scikit-learn`. Topic-number selection is based on model quality and interpretability, as discussed in the experimental setup, and we retain $K = 10$ topics as our preferred specification.

Each document is assigned a topic distribution, and we define the dominant topic as the one with the highest probability. These topic proportions provide a quantitative representation of campaign discourse.

To complement LDA, we also estimate a BERTopic model [4], which uses embedding-based representations and clustering methods to capture semantic similarities beyond word co-occurrence. This serves as a complementary check on topic interpretation.

4.2 Regression Analysis

To assess the determinants of political discourse, we regress topic proportions on candidate characteristics. Specifically, for each topic k, we estimate the following model:

$$Topic_{i,k} = \alpha + \beta\, Party_i + \gamma\, Profession_i + \delta\, Year_i + \varepsilon_i \tag{1}$$

where $Topic_{i,k}$ denotes the proportion of topic k in document i, while $Party_i$, $Profession_i$, and $Year_i$ capture party affiliation, candidate profession, and election year.

We estimate these models using ordinary least squares (OLS) with categorical encoding for party and profession. OLS is used here as a transparent linear framework for comparing the explanatory power of different metadata dimensions across topics. Our objective is descriptive rather than causal: we use these regressions to assess which characteristics are more strongly associated with variation in topic prevalence.

To quantify the relative importance of party and profession, we compute partial R^2 measures by comparing the full model with reduced models that exclude either *party* or *profession*. The change in explained variance provides a measure of the marginal contribution of each factor to topic usage.

4.3 Robustness Checks

We perform several robustness checks to assess the stability of our findings. First, we vary the number of topics in the LDA specification and re-estimate the regression models. Second, we test alternative preprocessing specifications, including more aggressive vocabulary filtering. Third, we re-run the analysis after excluding the largest residual party category ("Other") and after regrouping professions into broader categories.

Finally, we conduct a permutation test by randomly shuffling party labels and recomputing the regression results. This provides a baseline for evaluating whether the observed explanatory power of party affiliation could arise by chance. Across these alternative specifications, we examine whether the relative importance of party and profession remains stable.

4.4 Party-Level Analysis

In addition to regression analysis, we examine the structure of political discourse at the party level. We compute average topic distributions (centroids) for each party and measure the distance between individual documents and their party centroid in topic space. This provides an indicator of within-party thematic cohesion and dispersion.

We also compare parties in terms of topic dominance and average dispersion, providing complementary insights into how political organizations structure their messaging strategies. These analyses help contextualize the regression results and further highlight the role of party affiliation in shaping political discourse.

5 Experiments

This section describes the experimental setup used to extract topics and evaluate the determinants of political discourse. The analysis proceeds in two stages: topic modeling to represent campaign texts, and regression analysis to compare the explanatory importance of party affiliation and candidate profession.

5.1 Topic Modeling Setup

We construct a document-term matrix from the cleaned and lemmatized corpus using a bag-of-words representation using `CountVectorizer`. For specifications in `CountVectorizer`, tokens are filtered using a minimum document frequency of 20 and a maximum document frequency of 95%, and only words with at least three characters are retained.

We estimate Latent Dirichlet Allocation (LDA) models [1] with varying numbers of topics, $K \in \{5, 10, 15, 20, 25, 30\}$. Model quality is evaluated using several coherence metrics, including u_{mass}, c_v, c_{uci}, and c_{npmi}, together with perplexity. Lower-dimensional models achieve higher coherence, with $K = 5$ performing best on c_v. However, the corresponding topics remain broad and aggregate several political themes into large clusters. By contrast, although coherence is lower at K = 10, the resulting topics provide greater thematic differentiation and remain substantively interpretable. We therefore retain K = 10 as the main specification, as it offers a better balance between interpretability and analytical granularity for our research question.

Number of Topics (K)	c_v Coherence
5	0.600
10	0.562
15	0.558
20	0.548
25	0.536
30	0.519

Table 1: Topic coherence scores for different numbers of topics.

In addition to LDA, we estimate a BERTopic model [4] using multilingual sentence embeddings. This serves as a complementary robustness check for topic interpretation by capturing semantic similarities between documents beyond word co-occurrence.

5.2 Topic Representation

For each document, the LDA model produces a vector of topic proportions that sums to one. These vectors describe how strongly each document loads on each of the estimated topics. We define the dominant topic of a document as the topic with the highest probability.

These topic proportions are used as the main dependent variables in the regression analysis. In the preferred specification, this yields 10 topic variables capturing distinct dimensions of campaign discourse.

5.3 Regression Setup

To evaluate the determinants of topic usage, we estimate one ordinary least squares (OLS) model per topic, treating each topic proportion as a dependent variable. For each topic k, the explanatory variables include party affiliation, candidate profession, and election year.

To compare the relative importance of party and profession, we estimate a full specification and reduced specifications excluding either party or profession. The difference in R^2 between the full and reduced models is used as a partial R^2 measure of the marginal contribution of each factor.

We consider two year controls: a baseline specification in which year is treated as a categorical variable, and an alternative specification in which year is treated as a continuous variable. This allows us to assess whether the main results depend on how temporal variation is modeled.

5.4 Evaluation Metrics

Our evaluation focuses on two dimensions. First, we assess topic-model quality using coherence scores and perplexity. Second, we evaluate the explanatory power of the regression models using total R^2 and partial R^2 measures.

We also perform robustness checks across several dimensions: varying the number of topics, modifying preprocessing thresholds, excluding the dominant residual party category ("Other"), and regrouping professions into broader classes. Finally, we conduct a permutation test in which party labels are randomly shuffled to assess whether the observed party effect could arise by chance.

Implementation Details

All analyses are implemented in Python. Topic modeling is performed using `scikit-learn` for LDA and BERTopic for embedding-based topic extraction, while regression analyses are estimated with `statsmodels`. Experiments are run with a fixed random seed to ensure reproducibility.

6 Results

6.1 Topic Structure of the Corpus

The topic modeling stage reveals a structured and interpretable thematic space. The preferred LDA specification with $K = 10$ was selected as a balance between coherence and thematic granularity, and it identifies major dimensions of French electoral discourse, including ideological competition between left and right, social policy, labour and economic conflict, environmental politics, presidential and institutional politics, and national political rhetoric. BERTopic yields broadly similar semantic groupings, providing a complementary check on the interpretability of the LDA topics.

Table 2 summarizes the 10 topics using manual labels and representative terms from the preferred LDA specification.

Topic	Suggested label	Representative words
T1	Ideological competition	droite, communiste, gauche, politique, parti
T2	Social policy and welfare	social, france, emploi, formation, solidarité
T3	Labour and economic conflict	politique, gauche, travail, salaire, emploi
T4	Left-wing mobilization	communiste, majorité, changement, gauche, victoire
T5	Presidential and institutional politics	france, socialiste, président, député, république
T6	National political rhetoric	france, français, union, politique, national
T7	Economic and ecological policy	liberté, économie, écologiste, énergie, politique
T8	Right-wing electoral parties	national, front, vote, français, rpr
T9	Environmental politics	écologie, écologiste, environnement, verts, vie
T10	Party alliance and campaign rhetoric	entente, union, majorité, confiance, force

Table 2: Compact summary of the 10 LDA topics in the preferred specification. Topic labels are manual interpretations based on representative words.

At the document level, topic mixtures vary substantially across candidates, but clear regularities emerge once documents are aggregated by party. Some parties are highly concentrated on a narrow thematic profile, whereas others display more diversified topic mixtures. This pattern is consistent with the later regression and party-cohesion results, which show that party affiliation is the main organizing dimension of discourse in the corpus.

6.2 Party Versus Profession Effects

To quantify the determinants of topic usage, we estimate one regression per topic, using party affiliation, profession, and year as explanatory variables. Across all 10 topics, party affiliation contributes substantially more explanatory power than profession.

Using year as a categorical control, the average partial R^2 attributable to party is 0.407, compared with only 0.005 for profession. This gap appears for every topic. Party partial R^2 ranges from 0.134 to 0.739 across topics, whereas profession partial R^2 remains between 0.000 and 0.011. For example, party explains 0.696 of the variance in Topic 0 and 0.739 in Topic 8, while the corresponding profession effects are close to zero. Even in topics with lower total explained variance, party remains consistently more important than profession.

Topic	Total R^2	Party partial R^2	Profession partial R^2
Topic 0	0.864	0.696	0.002
Topic 1	0.554	0.475	0.003
Topic 2	0.320	0.222	0.009
Topic 3	0.527	0.295	0.009
Topic 4	0.507	0.335	0.009
Topic 5	0.321	0.256	0.011
Topic 6	0.185	0.134	0.004
Topic 7	0.482	0.372	0.003
Topic 8	0.977	0.739	0.000
Topic 9	0.870	0.547	0.000

Table 3: Explained variance in topic prevalence with year treated as a categorical control. Party affiliation explains substantially more variance than profession for all topics.

These findings remain essentially unchanged when year is treated as a numerical variable. In that specification, the average partial R^2 rises to 0.460 for party, while profession remains negligible at 0.005. Taken together, the regression results show that campaign themes are much more strongly associated with party affiliation than with candidate socio-professional background.

Mixed-Effects Specification

We also estimate mixed-effects models with year as a random intercept. These models confirm the same overall pattern: party effects remain substantively large, while profession effects are generally small. The estimated year-level random component is limited relative to the magnitude of the party coefficients, suggesting that temporal variation does not overturn the dominant role of party affiliation.

6.3 Party Cohesion and Dispersion

To better understand how parties structure discourse internally, we compute party centroids in topic space and measure the distance of each document to its party centroid. Figure 1 summarizes this structure by plotting each party's average discourse dispersion against its dominant-topic influence. Parties in the upper-left are more cohesive and more concentrated on a single theme, whereas those toward the lower-right are more diffuse.

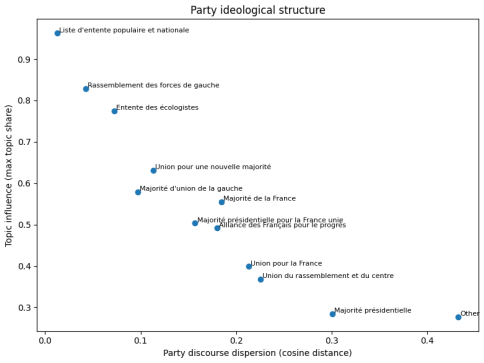

Figure 1: Party cohesion and topic concentration in topic space.

Clear differences emerge across parties. *Liste d'entente populaire et nationale* is the most cohesive party in topic space, with the lowest average distance to its centroid and the strongest dominant-topic concentration. *Rassemblement des forces de gauche* also appears highly cohesive. By contrast, broader categories such as *Other* are substantially more dispersed and much less concentrated around a single dominant topic. This complements the regression results by showing that party structures discourse not only across documents, but also within parties.

6.4 Robustness Checks

The main conclusion is robust to several alternative specifications. When the number of topics is varied, party remains much more important than profession, even though average party partial R^2 declines somewhat as the topic structure becomes more fragmented. Likewise, using more aggressive preprocessing thresholds still yields a strong party effect (0.363) and a negligible profession effect (0.005).

Two additional checks reinforce this conclusion. First, excluding the residual *Other* party category slightly increases the estimated role of party to 0.417, while reducing the profession effect to 0.003. Second, regrouping professions into broader classes produces the same qualitative result: party remains strongly explanatory at 0.424, while profession contributes only 0.002.

Finally, the permutation test provides strong evidence against a spurious party effect. When party labels are randomly shuffled, the mean party partial R^2 falls to approximately 0.003, far below the observed baseline value of 0.407. This indicates that the relationship between party affiliation and topic usage is not due to chance.

Specification	Party partial R^2	Profession partial R^2
Baseline	0.407	0.005
Different topic numbers	0.331	0.005
Different preprocessing	0.363	0.005
Exclude *Other* party	0.417	0.003
Grouped professions	0.424	0.002

Table 4: Robustness checks. Across all specifications, party remains far more explanatory than profession.

Summary of Findings

Taken together, the results consistently indicate that political discourse in this corpus is driven primarily by party affiliation. Profession has, at most, a marginal association with topic prevalence. The combination of topic modeling, regression analysis, party-level cohesion measures, and robustness checks points in the same direction: electoral messaging is overwhelmingly organized by party-level ideological structure.

7 Discussion

The central result of this study is that political discourse in the corpus is structured primarily by party affiliation rather than by candidate socio-professional background. Across regression specifications, party explains a substantial share of variation in topic prevalence, whereas profession contributes only marginally. This pattern remains stable across alternative topic numbers, preprocessing choices, profession regroupings, and sample restrictions. Taken together, these results suggest that campaign messaging in French legislative elections during this period was organized much more by collective political identities than by individual candidate profiles.

One interpretation is that parties act as strong coordinating institutions. Even though candidates differ in profession, age, and local context, they appear to speak within relatively stable partisan frames. This is consistent with party discipline, shared ideological repertoires, and strategic message coordination. In this perspective, campaign documents are not simply personal statements but expressions of broader partisan positioning.

At the same time, the results do not imply that candidate characteristics are irrelevant in all respects. Although profession does not substantially affect the thematic composition of campaign documents, it may still matter for candidate selection, local credibility, or rhetorical style. Moreover, the profession variable relies on broad grouping from highly granular raw metadata, which may attenuate subtler socio-professional differences. The weak profession effect should therefore be interpreted as evidence that profession is less important than party for topic composition in this corpus, not as evidence that professional identity plays no political role.

The party-dispersion analysis reinforces this interpretation. Several parties show strong thematic cohesion, with documents clustered closely around party centroids in topic space, whereas the residual category *Other* is much more dispersed. This suggests that some political organizations maintained especially disciplined campaign messages and strengthens the conclusion that party structure is a major source of discursive organization.

These findings also speak to debates on political personalization. For French legislative campaign materials from 1981, 1988 and 1993, discourse appears to remain strongly collectivized and party-centered. This does not rule out personalization in other settings, but it suggests that party remained the dominant level of organization in this historical context.

Several limitations should nevertheless be acknowledged. The corpus is affected by OCR noise, formatting artifacts, and uneven archival preservation, and the analysis covers only three election years. In addition, the profession variable had to be grouped into broad categories, which may attenuate finer socio-professional differences. More generally, the analysis is descriptive and associational rather than causal: party may operate through ideology, organizational control, candidate selection, or shared political environments, and identifying these mechanisms would require richer institutional or longitudinal data.

Despite these limitations, the results are substantively robust and methodologically informative. By combining thematic modeling with variance decomposition, measures of party-level cohesion, and permutation validation, this study offers a systematic approach to comparing the collective and individual determinants of political discourse. Overall, the evidence points to the predominance of party-level organization in shaping electoral messaging.

8 Conclusion

This study examined whether political discourse in French legislative campaign documents is structured primarily by party affiliation or by candidate socio-professional characteristics. Using a corpus of 3,596 campaign texts from the 1981, 1988, and 1993 elections, we combined topic modeling and regression analysis to study the determinants of campaign themes.

Our results show that party affiliation is by far the main factor associated with variation in political discourse. In the baseline specification, the average partial R^2 attributable to party is about 0.407, compared with only 0.005 for profession. This pattern remains stable across alternative topic specifications, preprocessing choices, profession groupings, and sample restrictions. Additional party-level analyses also show that several parties exhibit strong internal cohesion in topic space, reinforcing the interpretation that campaign messaging is largely organized at the party level.

Methodologically, this study illustrates the value of combining topic modeling with regression-based variance decomposition, party-level cohesion analysis, and robustness validation to study electoral communication. By linking document-topic distributions to candidate metadata, the study provides a systematic way to compare collective and individual determinants of political discourse.

These findings should nevertheless be interpreted with caution. The corpus is affected by OCR noise and archival constraints, the analysis covers only three election years, and the profession variable necessarily relies on broad categorization. Moreover, the results are associational rather than causal.

Overall, our findings suggest that French legislative campaign discourse during this period was shaped much more by party structure than by individual candidate background. Future work could extend this approach to other elections, more recent campaign materials, or alternative candidate-level characteristics in order to better understand how partisan coordination and political personalization interact over time.

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